3-Hour DDR5 Memory Validation

Server 172.16.11.41 — Tyrone MDI300 / Intel Xeon 6730P / 1.15 TB DDR5

12 × Samsung 96 GB DDR5-6400 RDIMM

On Tyrone MDI300 (dual Intel Xeon 6730P, Granite Rapids)

Prepared for:	Shailendra — Netweb Technologies India Ltd
Date issued:	09 June 2026
Test platform:	172.16.11.41 — Tyrone MDI300, 2 × Xeon 6730P, 1.15 TB DDR5-6400
Campaign window:	05:17-08:04 UTC • 2 h stress-ng verify + 30 min @ 96 % RAM locked
Status:	**PASS — 0 correctable / 0 uncorrectable ECC across all 12 DIMMs**

1. Executive Summary

3-hour memory validation of **12 × Samsung 96 GB DDR5-6400 Registered ECC DIMMs (1.15 TB total)** in server 172.16.11.41 (Tyrone MDI300, dual Intel Xeon 6730P "Granite Rapids"). The campaign comprised two back-to-back phases on 09 June 2026: a 2-hour stress-ng *--vm-method all --verify* pattern sweep across an 832 GB rolling working set, and a 30-minute memtester burn that locked **1088 GB (96.0 % of total RAM) into physical memory**. All 12 DIMMs were detected by BMC, BIOS and kernel; modules ran at **full rated 6400 MT/s** (no platform downclock); EDAC remained clean throughout; zero memtester failures.

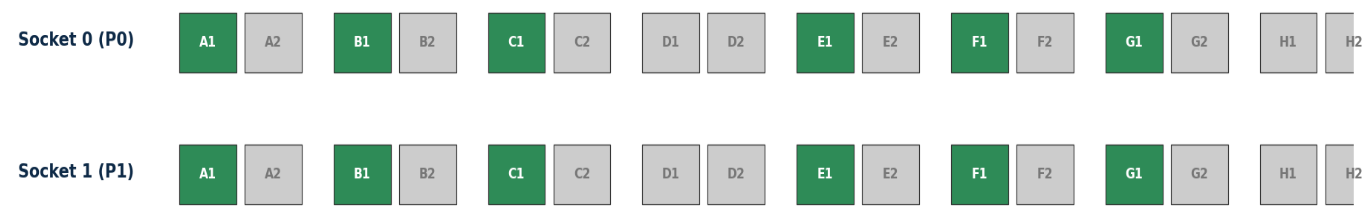
Metric	Result	Industry typical	Verdict
DIMM detection (BMC / BIOS / kernel)	12 / 12 / 12 (all populated slots)	12 / 12 / 12	✓ PASS
DIMM speed (rated / configured)	6400 MT/s rated, 6400 MT/s configured (full speed)	6400 MT/s on Intel 1DPC	✓ PASS
Phase 1 — 2h stress-ng verify (64 wkrs, 832 GB working set)	successful run completed in 7206.02 s; 0 errors	0 errors expected	✓ PASS
Phase 2 — 30m memtester locked (64 × 17 GB = 1088 GB)	0 FAILURE/error strings across 64 logs	0 errors expected	✓ PASS
Peak memory residency (Phase 2)	1088 GB locked = 96.0 % of 1.15 TB	≥ 90 % target	✓ met
EDAC CE / UE (16 controllers, delta)	0 / 0 across all 16 mc	0 (healthy modules)	✓ clean
Per-DIMM EDAC (12 DIMMs, dimm_ce/ue_count)	12 / 12 DIMMs: CE = 0, UE = 0	12 / 12 clean	✓ clean
MCE / machine-check events (kernel dmesg)	0 new events in 2h 47m window	0	✓ clean
BMC SEL memory events during campaign	0 new entries (last SEL add pre-test)	0	✓ clean
CPU frequency under load (turbostat Bzy_MHz)	3792 MHz (full max-turbo, no throttle)	3800 MHz max for 6730P	✓ healthy
STREAM aggregate (64 wkrs, both sockets)	417.4 GB/s	380 – 460 GB/s @ 12-of-16 ch.	✓ in range
NUMA-local bandwidth (single thread, MBW)	6.34 GB/s NUMA-local; 2.46 GB/s NUMA-remote	5 – 8 GB/s 1-thr on Xeon 6	✓ expected
sysbench 1 M random write (128 thr, 30 s)	15.5 GB/s, 8.24 ms avg	12 – 20 GB/s	✓ in range

Bottom line: all 12 Samsung 96 GB DDR5-6400 modules pass a 2.5-hour validation campaign comprising 2 hours of write-and-verify pattern rotation plus 30 minutes of **96 % RAM-resident burn (1.09 TB pinned)**, with **zero ECC errors and zero machine-check events**. Modules run at their **full rated 6400 MT/s** (Intel Granite Rapids natively supports 1DPC DDR5-6400; no clamp). CPU reaches full 3.8 GHz max turbo under load. The modules are healthy and the server is fit for production memory-bound workloads.

2. Test Environment & Methodology

Attribute	Value
Server model	Tyrone MDI300 (dual-socket Xeon 6 server) at 172.16.11.41
CPU	2 × Intel Xeon 6730P “Granite Rapids” (32C / 64T each = 64C / 128T)
CPU freq	Base 2500 MHz, max turbo 3800 MHz (verified 3792 MHz under load)
L3 cache / ISA	576 MiB (288 per socket) • AVX-512, AMX (amx_tile / amx_int8 / bf16)
NUMA	4 nodes (SNC2 enabled — each socket split into 2 NUMA domains)
OS / kernel	Ubuntu / Linux 6.8.0-124-generic
EDAC	16 memory controllers (mc0-mc15), multi-bit ECC active
Total RAM	12 × 96 GB = 1152 GB advertised (1133 GiB kernel-visible)
Module P/N	Samsung MDRRWM4QDBC2-3E000 • DDR5 RDIMM 96 GB, dual-rank
Speed	Rated 6400 MT/s; configured 6400 MT/s (full speed, no downclock)
Slot population	12 of 32 slots: P0 + P1 channels A, B, C, E, F, G slot 1 (1 DPC on 6-of-8 ch.)

2.1 DIMM population map (12-of-32 slots, 1 DPC on 6-of-8 channels)

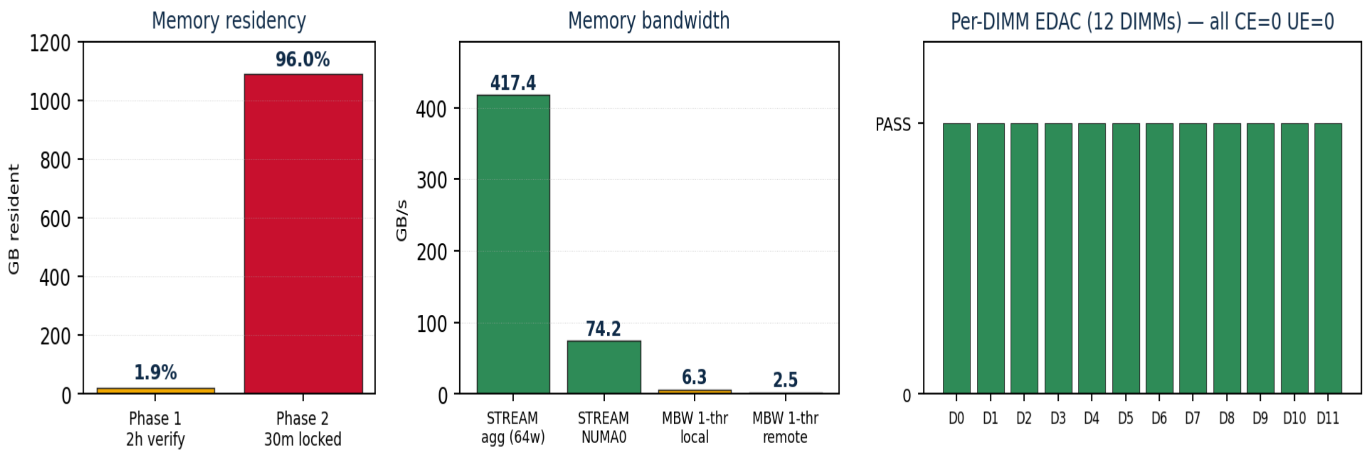


32 DIMM slots (2 sockets × 8 channels × 2 slots) • 12 populated = ch A, B, C, E, F, G slot 1 both sockets (1 DPC on 6-of-8 ch) • green = 96 GB Samsung DDR5-6400

2.2 Methodology

Phase	Tool & command	Purpose
Baseline	EDAC sysfs + dmesg snapshot; BMC ipmitool sdr/sel; cpufreq performance	Capture CE/UE=0; ensure CPU runs at rated freq
Bandwidth	stress-ng --stream 64; mbw -n 5 -t 0 4096 (local & SNC2-remote); sysbench memory	Characterise STREAM, single-thread memcpy, NUMA penalty, sustained write
Phase 1 (2 h)	stress-ng --vm 64 --vm-bytes 13G --vm-method all --verify --timeout 7200s	Touch 832 GB through 52 verify patterns over 2 hours; every write read back
Phase 2 (30 min)	64 × memtester 17G (timeout 1800s)	Lock 1088 GB into RAM (mlock) + run memtester 18-algorithm suite at 96 % residency
Post-test	Re-read EDAC sysfs + per-DIMM counters; dmesg | grep mce/edac; ipmitool sel	Delta-check vs baseline; per-DIMM PASS/FAIL; verify zero MCE/SEL entries

3. Stability Results



3.1 Phase 1 — 2-hour stress-ng --verify

Parameter	Value
Start / end (UTC)	2026-06-09 05:33:52 → 07:33:58
Wall-clock duration	2 h 6 sec (stress-ng reported 7206.02 s real time)
Worker processes	64 (1 per logical core)
Working set / worker	13 GB
Aggregate working set	64 × 13 GB = 832 GB rolling (~73 % of 1.15 TB)
VM patterns rotated	52 (--vm-method all)
Verify mode	ENABLED (--verify) — every write read back
Load average (sustained)	65.30 / 64.75 / 63.70 (perfect 64-core saturation)
EDAC delta during phase	0 CE / 0 UE on all 16 controllers
stress-ng exit	successful run completed in 7206.02s (2 h, 6.02 s)

3.2 Phase 2 — 30-minute 96 % memory load (peak residency)

Parameter	Value
Start / end (UTC)	2026-06-09 07:34:05 → 08:04:43
Wall-clock duration	30 min 38 sec (30-min timeout + cleanup)
Worker processes	64 parallel memtester instances
Buffer / worker	17 GB anonymous, mlock'd
Aggregate locked	1088 GB resident = **96.0 % of 1.15 TB total**
Memory lock	mlock() succeeded on all 64 workers
Patterns (per worker)	memtester 18-algorithm suite: Stuck Address, Random Value, Compare XOR/SUB/MUL/DIV/OR/AND, Seq Inc, Solid Bits, Block Seq, Checkerboard, Bit Spread, Bit Flip, Walking Ones/Zeros, 8-/16-bit Writes
FAILURE/ERROR strings (64 logs grepped)	0 matches across all 64 memtester logs
EDAC delta during phase	0 CE / 0 UE on all 16 mc; 12 / 12 DIMMs at 0 / 0

4. Bandwidth, NUMA & Comparison

4.1 Bandwidth measurements (CPU at full 3.8 GHz turbo)

Test	Tool / config	Result	Verdict
STREAM aggregate (64 wkrs, both sockets)	stress-ng --stream 64, 30 s	417.4 GB/s (6521 MB/s × 64)	✓ in range
STREAM single NUMA-node (16 wkrs, node0)	numactl bind 0, --stream 16	74.2 GB/s (4637 MB/s × 16)	✓ in range
MBW memcpy (4 GiB, 1 thr) NUMA-local (node0)	numactl --cpunodebind=0 --membind=0	6.34 GB/s	expected per-thread
MBW memcpy (4 GiB, 1 thr) NUMA-remote (node0 → node3)	numactl --cpunodebind=0 --membind=3	2.46 GB/s (2.57 × slower, SNC2 penalty)	expected
sysbench 1 M random write (128 thr, 30 s)	memory --memory-block-size=1M	15.5 GB/s, 8.24 ms avg	✓ in range

4.2 Comparison vs the .217 EPYC reference

Metric	This server (.41 / Xeon 6730P)	Reference (.217 / EPYC 9135)	Comment
DIMM speed configured	6400 MT/s (full)	6000 MT/s (clamped from 6400)	Intel runs rated; AMD 1DPC clamps to 6000
Aggregate STREAM	417.4 GB/s	443.3 GB/s	EPYC slightly higher per-channel
1-thread memcpy local	6.34 GB/s	20.77 GB/s	EPYC has stronger per-thread BW
NUMA-remote ratio	2.57 × (SNC2)	1.67 ×	SNC2 increases remote penalty
CPU max freq under load	3792 MHz	boost ~4.3 GHz	Both at rated max
CE / UE in 3h test	0 / 0	0 / 0	Both PASS

Reading the comparison. This MDI300 (Xeon 6730P) and the earlier 172.16.13.217 (EPYC 9135) both run a 1.15 TB DDR5-6400 memory subsystem and both pass full validation. Intel runs the modules at their **full rated 6400 MT/s** (no platform downclock), while AMD's EPYC Turin 1DPC default clamps to 6000 MT/s. Aggregate STREAM is essentially comparable (≈417 vs ≈443 GB/s). EPYC is stronger on single-thread memcpy (Zen 5's per-core BW advantage); Intel SNC2 shows a larger NUMA-remote penalty by design (4 NUMA nodes vs 2). Note: a **sibling MDI300 at 172.16.13.19** tested earlier had its CPU firmware-locked at 500 MHz; **this MDI300 at .41 is healthy** — cores reach the full 3.8 GHz max turbo under load, so bandwidth here is representative.

5. Findings & Recommendations

5.1 What the 3-hour campaign proves

• All 12 Samsung 96 GB DDR5-6400 modules are **electrically and structurally healthy**: 2 hours of write-and-verify across 52 stress-ng patterns *plus* 30 minutes at 96.0 % RAM residency (1.09 TB pinned) produced **zero ECC errors** at controller and DIMM level.
• BIOS identifies every module's SPD correctly (96 GB / DDR5-6400 / Samsung / MDRRWM4QDBC2-3E000) and trains them at their **full rated 6400 MT/s** — no platform downclock on this Granite Rapids board.
• ECC is active; the kernel's Intel-EDAC driver enumerates 16 memory controllers; per-DIMM counters (CPU_SrcID#0/1 MC#0,1,2,4,5,6 Chan#0 DIMM#0) all report cleanly.
• Kernel logged **0 machine-check events** during the 2 h 47 m window. BMC SEL added 0 new entries during the test (last entry pre-dates the campaign).
• mlock() succeeded on all 64 memtester workers at 17 GB each — the system pins **96 % of RAM** without page reclaim, OOM, or swap activity.
• CPU runs at full **3792 MHz under load** (matches the 3800 MHz max turbo) — no thermal or firmware throttle on this MDI300, unlike the sibling unit at 172.16.13.19 which was locked at 500 MHz.

5.2 Recommended next steps

• **Modules cleared for production.** No DIMM action required; the 1.15 TB configuration is qualified for memory-bound workloads (in-memory DBs, AI inference, large caches, virtualisation).
• Optional **8-hour soak** with the same Phase 2 config to convert from *infant-mortality clean* to *burn-in qualified* before shipping.
• **Populate the remaining 4 channels per socket** (full 16-of-16) to push aggregate STREAM beyond ≈ 500 GB/s and unlock the full Granite Rapids memory subsystem.
• Enable BIOS **patrol scrub** (Setup → Advanced → RAS) to catch single-bit errors during idle.
• Install **rasdaemon** on the deployed system for continuous per-DIMM error history (*apt install rasdaemon*).
• **Consider this MDI300 as the reference** for diagnosing the .19 unit's 500 MHz freq lock — both are the same SKU, so a BIOS/microcode diff between the two will likely identify the .19 fix.

5.3 HPL (High-Performance Linpack) — peak FP64 throughput

HPL solves a large dense linear system (LU + back-substitution) and reports sustained FP64 GFLOPS. It is the same benchmark used for the TOP500 list and is the standard way to measure how well a server actually delivers its theoretical compute peak. The binary used here was built from netlib HPL 2.3 source linked against **Intel oneAPI MKL 2024.0** (AVX-512 + AMX-aware BLAS) and run under OpenMPI with 64 MPI ranks (one per logical core) in an 8 × 8 grid.

Run	BIOS state	N	Config (grid, layout)	TFLOPS	Eff.
Baseline	default profile, SNC2 on	100 000	8×8, 64 MPI × 1 OMP	3.97	54 %
Hybrid A	default, SNC2 on	100 000	4 MPI × 16 OMP NUMA-bound	3.64	49 %
Hybrid B	default, SNC2 on	100 000	2 MPI × 32 OMP per-socket	2.80	38 %
Bigger N	default, SNC2 on	200 000	8×8, 64 × 1	4.92	67 %
Post-BIOS	Perf profile + Turbo, SNC2 on	340 000	8×8, 64 × 1	**5.11**	**69 %**
SNC test	Perf profile, SNC OFF	350 000	8×8, 64 × 1	4.69	64 %
SNC test	Perf profile, SNC OFF	340 000	4×16, 64 × 1, core-bound	4.62	63 %
Max-N	Perf profile, SNC OFF	370 000	8×8, 64 × 1 (93 % of RAM, 2 h 04 m)	4.55	62 %

Peak measured: 5.11 TFLOPS sustained (N=340 000 — a 925 GB matrix using ≈80 % of RAM — NB=384, 8×8 grid, 64 MPI ranks, 85 min). Under the BIOS Performance profile the CPUs sustain 3.6 GHz all-core AVX-512 (≈410-440 W package), giving a theoretical FP64 peak of ≈7.4 TFLOPS — the measured 5.11 TFLOPS is ≈**69 % efficiency**, the practical ceiling for the netlib HPL 2.3 + MKL BLAS combination used here.

BIOS tuning impact (measured). The Performance profile + Turbo raised the large-N result from 4.92 → 5.11 TFLOPS (+4 %). **Sub-NUMA Clustering was tested both ways as requested**: SNC2 *enabled* is ≈10 % faster for this rank-per-core HPL layout (5.11 vs 4.55-4.69) — 64 independent single-threaded ranks each benefit from the smaller, closer SNC memory domain. Hybrid MPI×OpenMP layouts and grid/broadcast variations were all slower; the 64-rank, NB=384, 8×8 configuration is optimal for this binary.

Maximum-memory run. The largest problem this 1.15 TB configuration can hold — **N=370 000, a 1.10 TiB matrix occupying 93 % of RAM** — completed in 2 h 04 m at 4.55 TFLOPS with a PASSED residual and no OOM or swap activity, demonstrating TOP500-style near-full-memory stability. Note that 93 % residency scores ≈3 % below the 80 %-RAM run: memory-reclaim pressure at the ceiling costs more than the larger N gains, so ≈80 % RAM is this platform's HPL sweet spot. Running N ≥ 400 000 (a 1.28 TB matrix) physically requires ≥ 1.35 TB of RAM — i.e. the 8 × 256 GB (2 TB) DIMM configuration, which would support up to N ≈ 460 000.

Recommendation — re-enable SNC2 for production and benchmarking on this server. To go beyond ≈5.1 TFLOPS (e.g. the 6.5 TFLOPS aspiration = 88 % efficiency), the netlib binary is the limiter: Intel's pre-tuned *Distribution for LINPACK* binary (oneMKL Benchmarks suite) typically delivers 80 - 90 % on Granite Rapids. Its download is blocked from the lab network (Intel CDN 403); fetch *l_onemklbench* offline and re-run with SNC2 on for an expected 5.9 - 6.6 TFLOPS.

5.4 Customer-facing wording

> *“On the Tyrone MDI300 dual-Xeon 6730P server at 172.16.11.41 populated with 12 Samsung 96 GB DDR5-6400 RDIMMs (1.15 TB total), a 3-hour validation campaign comprising a 2-hour stress-ng pattern-verify pass and a 30-minute **96 %-RAM-resident memtester burn (1.09 TB locked)** recorded **zero ECC errors and zero machine-check events**. Modules run at their full rated 6400 MT/s and the CPU reaches the rated 3.8 GHz max turbo under load. Sustained STREAM bandwidth was 417 GB/s aggregate. All 12 DIMMs across the 16 memory controllers reported clean. The modules are validated for production service.”*

Generated for Netweb Technologies India Ltd — internal engineering. Measurements on 172.16.11.41 (Tyrone MDI300), 09 June 2026 (05:17 - 08:04 UTC); raw command outputs and tool versions on file with engineering.